

CHRISTOPHER E. H. LUTSKO

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October 7, 2026

BIOGRAPHICAL

Born: April 26 1994

Pronouns: He/Him

Citizen: USA/UK

Spoken Languages: English, French, Spanish, German.

CAREER

University of Houston

2024 +

Assistant professor

University of Zürich

2023-2024

Postdoctoral researcher

Rutgers University

2020-2023

Hill assistant professor

EDUCATION

University of Bristol

2016-2020

PhD in Mathematics

Advisors: Jens Marklof & Bálint Tóth

University of Texas at Austin

2012-2016

B.S Mathematics

Dean's Scholars Honors Program

PUBLICATIONS

Preprints:

28. *Spacing statistics for a point scatterer on the cubic three-torus*, arXiv:2610.10031 (2026).

27. *Spherical density for real semisimple groups*, arXiv:2610.00298 (2026).

26. *Loss of memory in the periodic Ehrenfest model with small polyhedral scatterers*
with J. Marklof, arXiv:2609.04371 (2026).

25. *Poissonian pair correlation for the three-particle Sutherland model*, arXiv:2609.03112 (2026).

24. *Improved sup-norm bounds for locally symmetric spaces*, arXiv:2608.21580 (2026).

23. *Poissonian correlations of $\alpha n^d \bmod 1$*
with N. Rome and N. Technau, arXiv:2605.06974 (2026).

22. *The Gauss circle problem for Penrose tilings*
with A. Haynes, arXiv:2512.21444 (2025).

21. *Average variance bounds for integer points on the sphere*, arXiv:2402.12822 (2024).

Papers:

20. *Exceptional eigenvalue density for thin groups*,
Int. Math. Res. Not. (IMRN), **2026**(18), 208 (2026).
19. *Sign changes along geodesics of modular forms*
with D. Kelmer and A. Kontorovich (2024) [Accepted: **J. Théor. Nombres Bordeaux**].
18. *Counting in lattice orbits*
with A. Kontorovich (2024) [Accepted: **Bull. London Math. Soc.**].
17. *Diffusion of the random Lorentz process in a magnetic field*
with B. Tóth, **J. Math. Phys.**, **66**(11), (2025) (Editor's pick).
16. *Polyhedral bounds on the joint spectrum and temperedness of locally symmetric spaces*
with T. Weich and L. Wolf (2024) [Accepted: **Duke Math. J.**].
15. *An abstract spectral approach to horospherical equidistribution*,
Nonlinearity, **38**(10), 105014, (2025).
14. *Hyperbolic lattice point counting in unbounded rank*
with V. Blomer, **J. Reine Angew. Math.**, **2024**(812), 257-274, (2024).
13. *Norm bounds on Eisenstein series*
with D. Kelmer and A. Kontorovich, **Int. J. Number Theory**, **20** (08), 2083-2098 (2024).
12. *These Numbers Look Random but Aren't, Mathematicians Prove*
Scientific American (2024) (also featured in **Spektrum der Wissenschaft**).
11. *m-Point Correlations of the Fractional Parts of αn^θ*
with N. Technau, **Amer. J. Math.**, **148**(2), 569-597 (2026).
10. *Full poissonian local statistics of slowly growing sequences*
with N. Technau, **Compos. Math.**, **161**(1), 148-180, (2025).
9. *Effective counting in sphere packings*
with A. Kontorovich, **J. of the Assoc. Math. Res.**, **2**, 15-52 (2024).
8. *Sarnak's spectral gap question*
with D. Kelmer and A. Kontorovich, **J. Anal. Math.** **151**, 171-179 (2023) (special volume for P. Sarnak).
7. *Pair correlation of the fractional parts of αn^θ*
with A. Sourmelidis and N. Technau, **J. of the Eur. Math. Soc. (JEMS)**, **27**(10), 4069-4082 (2024).
6. *Long-range correlations of sequences modulo 1*,
J. of Number Theory, **234**, 333-348 (2022).
5. *Farey sequences for thin groups*
Int. Math. Res. Not. (IMRN), **15**, 11642-11689 (2022).
4. *Invariance principle for the random wind-tree process*
with B. Tóth, **Ann. Henri Poincaré**, **22**(10), 3357-3389 (2021).
3. *Invariance principle for the random Lorentz gas – beyond the Boltzmann-Grad limit*
with B. Tóth, **Comm. in Math. Phys.**, **379**, 589-632, (2020).
2. *Directions in orbits of geometrically finite hyperbolic subgroups*,
Math. Proc. of the Cambridge Philos. Soc. **171**(2), 277-316 (2021).

1. *A microscopic approach to nonlinear Reaction-Diffusion: the case of morphogen gradient formation*
with J.P. Boon, J.F. Lutsko, **Phys. Rev. E** 85:021126 (2016).

PhD Thesis:

Statistical Properties of Dynamical Systems: From Statistical Mechanics to Hyperbolic Geometry, 2020 (Bristol University). [Nominated for doctoral dissertation prize]

Conference Proceedings:

1. *Invariance principle for random Lorentz gas in the Boltzmann-Grad Limit*, Oberwolfach Report 10/2019 p. 33-35 (2019)
2. *Invariance principle for random Lorentz gas — Beyond the Boltzmann-Grad Limit*, Oberwolfach Report 42/2019 p. 12-15 (2019)

FUNDING AND AWARDS

Simons Travel Grant (5 years, \$35,000)	2025
Nominated for doctoral dissertation prize for outstanding excellence	2020
William's Scholar Honored Graduate	2016

TEACHING

Instructor:

Houston: Math 6322 Functions of Complex Variables	Fall 2026
Houston: Math 2451 Accelerated Calculus	Spring 2026
Houston: Math 3338 Probability	Fall 2025
Houston: Math 2451 Accelerated Calculus	Spring 2025
Houston: Math 2450 Accelerated Calculus	Fall 2024
Rutgers: Math 152 Calculus II (Math/Physics)	Spring 2023
Rutgers: Math 152 Calculus II (Math/Physics)	Spring 2023
Rutgers: Math 356 Theory of Numbers	Fall 2022
Rutgers: Math 437 History of Mathematics	Spring 2022
Rutgers: Math 152 Calculus II (Math/Physics)	Spring 2022
Rutgers: Math 356 Theory of Numbers	Fall 2021
Rutgers: Math 356 Theory of Numbers	Spring 2021
Rutgers: Math 152 Calculus II (Math/Physics)	Spring 2021
Rutgers: Math 152 Calculus II (Math/Physics)	Fall 2020

Teaching Assistant:

Acted as TA for approximately 11 different courses from 2013-2024

POST-DOCS

Kavita Dhanda	2026+
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GRADUATE STUDENTS

Anish Ray	2025+
Ziyue Song	2026+
Thomas Zhang	2026+

UNDERGRADUATE MENTORSHIP

Arin Morantes (Undergraduate thesis advisor)

Summer 2026+

ORGANIZATIONAL DUTIES

Linfoot number theory seminar (Bristol)

Fall 2019 - Spring 2020

Rutgers number theory seminar

Fall 2021 - Spring 2023

Houston dynamics seminar

Fall 2024 +

Rice-UH Dynamics Day (once per semester)

Spring 2026 +

ACADEMIC INVITATIONS

1 week - Budapest University of Technology (BME)

2017

2 weeks - University of Bonn

2023

1 week - UW Madison

2026

SEMINAR TALKS

University of Bristol

2017

Budapest University of Technology

2017

University of Bristol

2017

University of Warwick

2018

University of Bristol

2018

University of Bristol

2018

Univ. Libre de Bruxelles

2018

University of Bristol

2018

Rutgers University

2019

University of Texas at Austin

2019

University of Houston

2019

University of Exeter

2019

Manchester University

2019

University of Bristol

2019

University of Loughborough

2020

University of Oklahoma

2020

Yeshiva University

2020

Rutgers University

2021

Tata Institute of Fundamental Research

2021

New England Dynamics and Number Theory Seminar

2021

(recording: <https://www.youtube.com/watch?v=EODTepggUuU>)

Hong Kong University

2022

University of Illinois Urbana-Champaign

2022

Stony Brook **Colloquium**

2022

(recording: <https://www.youtube.com/watch?v=SOHhKdno2jA>)

Rutgers University

2022

New England Dynamics and Number Theory Seminar

2022

(recording: <https://nednt.wescreates.wesleyan.edu/fall-2022-schedule/>)

University of Maryland

2022

Philadelphia Area Number Theory Seminar (Bryn Mawr)

2023

Rutgers University

2023

University of Bonn

2023

Paderborn University

2023

University of Zurich

2023

EPFL

2024

University of Bristol	2024
Manchester University ($\times 2$)	2024
University of Houston Colloquium	2024
Rice University	2024
One World Probability Seminar	2024
Yale University	2024
University of Houston	2025
IST Austria - University of Graz Joint Seminar	2025
UW Madison	2026
International Seminar on Automorphic Forms	2026
(recording: https://www.youtube.com/watch?v=PuWv7AB5Tgg)	
Rice University	2026

CONFERENCE TALKS

MINGLE post-graduate event - University of Bristol, UK	2017
Dynamics Days Europe (Billiards Minisymposium) - University of Loughborough, UK	2018
Probability and NonLocal PDEs - University of Swansea, UK	2018
Mini-Workshop: Lorentz Gas Dynamics - M. F. Oberwolfach, Germany	2019
Large Scale Stochastic Dynamics - M. F. Oberwolfach, Germany	2019
MINGLE post-graduate event - University of Bristol, UK	2019
CRC Workshop "Geometry and analysis of locally symmetric spaces" - Paderborn, Germany	2024
Stochastics and Influences - Erdős Center, Budapest, Hungary	2025
Rice-UH Dynamics Day	2026
BIRS Workshop, "Arithmetic, L-functions, and Pseudorandomness" - Banff, Canada	2026

OUTREACH

Volunteer Chemistry Tutor – Garza high school (Austin, Texas)	2015
Help with Bristol maths day	2018
Participant in "Skype a Scientist"	2021
Speaker at Chicago public schools teen mentoring program STEM event	2021
Contributor to EMS magazine problems section	2022
Help with Museum of mathematics' polyplane project	2023
Contributor to Scientific American [link] (this article was also featured in Spektrum der Wissenschaft)	2024
Interview on ABC 13 Houston (Houston area television news program)	2025

REFEREEING/REVIEWING/SERVICE

AMS Mathematical Reviews/MathSciNet [12 reviews] • Advances in Mathematics • American Journal of Mathematics • Annales Henri Poincaré • Chaos, Solitons and Fractals • Communications of the AMS • Compositio Mathematica • Ergodic Theory and Dynamical Systems • Indagationes Mathematicae • International Mathematics Research Notices • Journal of the European Mathematical Society • Journal of the London Mathematical Society • Journal of Mathematical Physics • Journal of Number Theory • Journal of Statistical Physics • Mathematical Proceedings of the Cambridge Philosophical Society • Nonlinearity • Random Structures and Algorithms • Research in Number Theory • Stochastic Processes and their Applications